

WASP-43b

R-1582

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_210311 · created 2026-08-02T00:01:10+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459284.6804 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1532 +/- 0.009
Transit depth (Rp/Rs)^2	2.35 +/- 0.27 [%]
Orbital Inclination (inc)	87.1 +/- 1.04
Airmass coefficient 1 (a1)	0.955 +/- 0.0096
Airmass coefficient 2 (a2)	0.0273 +/- 0.0036
Scatter in the residuals of the lightcurve fit is	1.01 %
Best Comparison Star	#2 - [577, 282]
Optimal Method	PSF photometry
Transit Duration (day)	0.0601 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1532 ± 0.009 vs 0.1594 (deviation 0.68σ)
Duration fit vs published	0.0601 d vs 0.0512 d (deviation 7.89σ)
Mid-time O–C	2.8 min
Depth SNR	16.7
Residual scatter	1.01 %

Light curve

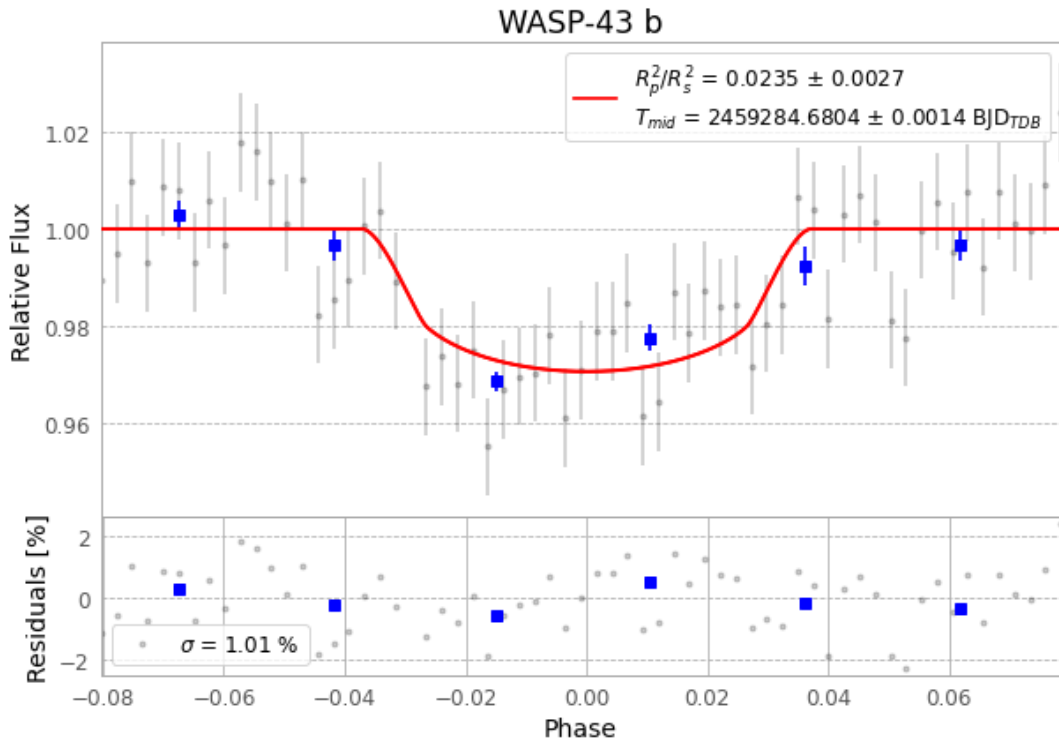


Figure 1 — FinalLightCurve_WASP-43 b_2021-03-10.png (SHA-256 f3f36b45c844fe22...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [491, 234], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3c71c2e08f195b8a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), WASP-43210311023612.FITS → WASP-43210311054212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-210311.log (bf0a64a5d094...), FinalLightCurve_WASP-43 b_2021-03-10.png (f3f36b45c844...), FinalLightCurve_WASP-43 b_2021-03-10.csv (41fac04e98ce...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)